

Summary Report

Approach:

I approached this project as if I were building an early prototype for a real hospital analytics pipeline. I began with structured patient data, cleaned and standardized all fields, and then engineered features to better capture clinical patterns converting categorical values, scaling numeric ones, and integrating discharge note text through TF-IDF.

I trained and evaluated three models - Logistic Regression, Random Forest, and XGBoost to predict 30-day readmission. The entire workflow was built in reproducible pipelines, ensuring any future data can flow through the same process.

For the unstructured text, I used spaCy with custom rule patterns to extract diagnoses, medications, symptoms, treatments, and follow-up actions. These extractions were quantified and merged back into the structured dataset to see how clinical notes could add signal to readmission prediction.

Key Results:

Among all models, XGBoost delivered the most balanced performance (ROC-AUC ≈ 0.51 , F1 ≈ 0.53). While modest, this improvement shows the potential of text-driven features in identifying risk signals that structured data alone might miss. The NLP pipeline also proved valuable it successfully transformed messy clinical notes into interpretable entities that can be audited, visualized, or fed into downstream models.

Practical Implications:

Even this lightweight prototype demonstrates how combining structured EHR data with free-text insights can give care teams an early read on high-risk discharges. In a production setting, such a system could help prioritize follow-ups, reduce avoidable readmissions, and provide a transparent audit trail for clinical review.

If I Had More Time or Data:

With more data, I would fine-tune domain-specific models like ClinicalBERT or SciSpaCy to capture medical context and negation (“no chest pain”) more accurately. I’d also test time-aware features (length of stay, prior admissions) and explore explainability tools such as SHAP for clinician-friendly model insights. Finally, I’d integrate clinician feedback loops to continuously refine both the prediction and the entity extraction components, turning this prototype into a reliable decision-support tool.